

A determinant difference bound in the sup operator norm

MathlibAnnex.ContinuousLinearMap.abs_det_sub_le_max

theorem

Controls determinant variation by replacing one matrix row at a time.

Statement

Let I be a finite set of size n , and give $\mathbb{R}^I$ the sup norm $\|x\|_\infty = \max_{i \in I} |x_i|$ (the zero norm on the zero space if $I = \emptyset$). For continuous real-linear maps $P, Q : \mathbb{R}^I \rightarrow \mathbb{R}^I$, use the induced operator norm. If $n \geq 1$, then

$$|\det P - \det Q| \leq n \max\{\|P\|, \|Q\|\}^{n-1} \|P - Q\|.$$

For $n = 0$, both determinants are 1 and the bound is $0 \leq 0$.

Assumptions

The coordinate set I is finite and may be empty. The coefficients are real and the operator norm is induced by the sup norm on both domain and codomain. Neither map is assumed invertible.

Conclusion

The coefficient n counts the number of row replacements. The norm is specifically the sup operator norm; the estimate is not being asserted here with an unspecified norm on the same coordinate space.

The row estimate uses the sum of the absolute values of entries, not the Euclidean length of a row. This fits the sup norm through a sign vector. All coordinate matrices refer to the standard coordinate basis, so their determinants are the determinants of the original linear maps.

Proof route

The determinant is bounded by the product of the absolute row sums. A sign vector bounds each such row sum by the sup operator norm. In a one-row replacement, one row is a row of $P - Q$ and the other $n - 1$ rows are rows of P or Q ; summing the n replacement bounds gives the result.

▼ Proof steps

1. **Bound a determinant by its row sums.** Let $C = (C_{ij})_{i,j \in I}$ be a real square matrix, and let $\mathfrak{S}(I)$ denote the permutations of I . Expanding the determinant gives

$$\begin{aligned}
|\det C| &= \left| \sum_{\sigma \in \mathfrak{S}(I)} \operatorname{sgn}(\sigma) \prod_{i \in I} C_{i, \sigma(i)} \right| \\
&\leq \sum_{\sigma \in \mathfrak{S}(I)} \prod_{i \in I} |C_{i, \sigma(i)}| \\
&\leq \sum_{f: I \rightarrow I} \prod_{i \in I} |C_{i, f(i)}| \\
&= \prod_{i \in I} \left(\sum_{j \in I} |C_{ij}| \right).
\end{aligned}$$

The second inequality includes all functions $f: I \rightarrow I$, rather than only bijections; every added summand is nonnegative. The last equality is distributivity: expanding the product chooses one column $f(i)$ independently in each row. This is precisely the row-sum bound needed below, not a bound by Euclidean row lengths.

2. Bound each row sum by the operator norm. Let C now be the standard matrix of a continuous real-linear map $T: \mathbb{R}^I \rightarrow \mathbb{R}^I$, so $C_{ij} = (Te_j)_i$ for the coordinate vector e_j . For a fixed row i , choose

$$\varepsilon_j^{(i)} = \begin{cases} 1, & C_{ij} \geq 0, \\ -1, & C_{ij} < 0. \end{cases} \quad \|\varepsilon^{(i)}\|_\infty \leq 1.$$

Then $C_{ij}\varepsilon_j^{(i)} = |C_{ij}|$, and therefore

$$\begin{aligned}
\sum_{j \in I} |C_{ij}| &= |(T\varepsilon^{(i)})_i| \\
&\leq \|T\varepsilon^{(i)}\|_\infty \leq \|T\| \|\varepsilon^{(i)}\|_\infty \leq \|T\|.
\end{aligned}$$

Apply this with $T = P$, $T = Q$, and $T = P - Q$. These three applications provide the bounds for every row appearing in the next step.

3. Estimate one row replacement. Assume $n \geq 1$ and choose one enumeration of I for both rows and columns. Write $P_{ij} = (Pe_j)_i$ and $Q_{ij} = (Qe_j)_i$. Define

$$(C_k)_{ij} = \begin{cases} P_{ij}, & i \leq k, \\ Q_{ij}, & i > k, \end{cases} \quad 0 \leq k \leq n.$$

Thus C_0 is the matrix of Q and C_n that of P . For $1 \leq k \leq n$, let D_k have row k equal to $(P - Q)_{k, \bullet}$ and every other row equal to that of C_{k-1} . Linearity of the determinant in row k gives

$$\det C_k - \det C_{k-1} = \det D_k.$$

Step 2 bounds the row sums of D_k by

$$\sum_j |(D_k)_{ij}| \leq \begin{cases} \|P - Q\|, & i = k, \\ \max\{\|P\|, \|Q\|\}, & i \neq k. \end{cases}$$

Substituting these bounds into Step 1 yields

$$|\det C_k - \det C_{k-1}| \leq \max\{\|P\|, \|Q\|\}^{n-1} \|P - Q\|.$$

There is one difference row and exactly $n - 1$ other rows. For $n = 1$, the product over the other rows is the empty product 1.

4. **Sum the n replacement bounds.** Since $C_0 = Q$ and $C_n = P$ in the standard coordinates,

$$\begin{aligned} |\det P - \det Q| &= \left| \sum_{k=1}^n (\det C_k - \det C_{k-1}) \right| \\ &\leq \sum_{k=1}^n |\det C_k - \det C_{k-1}| \\ &\leq n \max\{\|P\|, \|Q\|\}^{n-1} \|P - Q\|. \end{aligned}$$

This is the stated bound. If $n = 0$, there are no replacements and both empty determinants are 1, giving $0 \leq 0$ separately.

Main citations

- Exact declaration and its source — `MathlibAnnex.ContinuousLinearMap.abs_det_sub_le_max`
- A sign vector bounds each matrix row by the operator norm — `MathlibAnnex.ContinuousLinearMap.rowL1_stdMatrix_le_opNorm`
- One-row determinant difference — `MathlibAnnex.ContinuousLinearMap.det_rowsFrom_insert_sub`
- The one-row estimate and its finite-set sum — `MathlibAnnex.ContinuousLinearMap.abs_det_rowsFrom_sub_le_max`
- Standard matrix and linear-map determinants agree — `MathlibAnnex.ContinuousLinearMap.det_stdMatrix`
- The sum of absolute values in a row — `MathlibAnnex.Matrix.rowL1Norm`
- The determinant bound by row sums — `MathlibAnnex.Matrix.abs_det_le_prod_rowL1Norm`

Lean source signature (exact)

```
theorem abs_det_sub_le_max
  {ι : Type u} [Fintype ι] [DecidableEq ι]
  (P Q : (ι → ℝ) →L[ℝ] (ι → ℝ)) :
  |LinearMap.det (P : (ι → ℝ) →ι[ℝ] (ι → ℝ)) -
   LinearMap.det (Q : (ι → ℝ) →ι[ℝ] (ι → ℝ))| ≤
  (Fintype.card ι : ℝ) *
  (max ‖P‖ ‖Q‖) ^ (Fintype.card ι - 1) * ‖P - Q‖
```

In the source	Mathematical meaning
<code>{ι : Type u} [Fintype ι] [DecidableEq ι]</code>	The finite coordinate set I of size n . Equality decisions support finite-index operations, rather than restrict the maps geometrically.
<code>(P Q : (ι → ℝ) →L[ℝ] (ι → ℝ))</code>	The continuous real-linear endomorphisms P, Q of $\mathbb{R}^I$ with its sup norm. Their norms are the induced operator norms.
<code>LinearMap.det (P : (ι → ℝ) →_ι[ℝ] (ι → ℝ))</code>	$\det P$, obtained by forgetting the continuity certificate without changing the map. The analogous expression gives $\det Q$.

In the source	Mathematical meaning
<code>(Fintype.card ι : ℝ)</code>	The real number $n = I $ counting the rows.
<code>(max ‖P‖ ‖Q‖) ^ (Fintype.card ι - 1)</code>	For $n \geq 1$, $\max\{\ P\ , \ Q\ \}^{n-1}$. Natural-number subtraction makes the exponent 0 at $n = 0$.

In the source	Mathematical meaning
<code>‖P - Q‖</code>	The sup operator norm of the difference $P - Q$.
<code> LinearMap.det (P : (ι → ℝ) →_l [ℝ] (ι → ℝ)) - LinearMap.det (Q : (ι → ℝ) →_l [ℝ] (ι → ℝ)) ≤ (Fintype.card ι : ℝ) * (max ‖P‖ ‖Q‖) ^ (Fintype.card ι - 1) * ‖P - Q‖</code>	The displayed full inequality is $ \det P - \det Q \leq n \max\{\ P\ , \ Q\ \}^{n-1} \ P - Q\ $ when $n \geq 1$; at $n = 0$ it reads $0 \leq 0$.

▼ Exact surrounding binder context (separate excerpt)

```

noncomputable section

open Equiv Finset
open scoped BigOperators

namespace MathlibAnnex

```

▼ Exact content identity

Declaration: `MathlibAnnex.ContinuousLinearMap.abs_det_sub_le_max`
Accepted content SHA-256: `5f4d67395dbe0ba5df4b29d5df0eff5476d758d834bc02cb9031efe32d276f5c`
Accepted source guide SHA-256: `f3fbd6ce3f86e42fde4e741cc1984c4d33829563ebc554ed37cd7c38217db202`
Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`